Problem & Project Definition



University of Ruhuna

Department of Information and Communication Technology

HomeLink

A Home Services Booking Platform

Project Proposal

Course: ICT2232 - Software Engineering Mini Project

Prepared By:	
Project Type:	
Target Services:	

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A Centralized Home Services Booking Platform

1. Background / Problem Domain

The demand for home services such as electrical repairs, plumbing, cleaning, and other household maintenance services is common among households. However, finding a suitable service provider and arranging a service can sometimes be inconvenient. Customers may have to depend on personal contacts, recommendations, or different communication methods to find available service providers and arrange bookings.

At the same time, individual service providers may have limited ways to make their services visible to potential customers and manage incoming service requests in an organized manner. The lack of a centralized platform can make the overall process of connecting customers and service providers less convenient for both parties.

HomeLink is proposed as a Home Services Booking Platform that provides a centralized system for connecting customers with home-service providers such as electricians, plumbers, and cleaners. Through the platform, customers will be able to find suitable service providers and arrange and manage service bookings, while service providers will be able to present their services and manage customer service requests.

2. Problem Statement

People who require home services such as electrical repairs, plumbing, cleaning, and other household maintenance may face difficulties in finding suitable service providers and arranging services conveniently. The process often depends on personal contacts, recommendations, or separate communication methods, which can make finding and coordinating services inconvenient.

At the same time, home-service providers may have limited opportunities to make their services visible to potential customers and manage service requests in an organized manner. Therefore, there is a need for a centralized platform that can efficiently connect customers who require home services with suitable service providers and support the process of arranging and managing service bookings.

3. Stakeholders

The main stakeholders of HomeLink are classified as primary stakeholders and secondary stakeholders.

3.1 Primary Stakeholders

Customers
Customers are people who require home services such as electrical repairs, plumbing, cleaning, and other household maintenance services. They need a convenient way to find suitable service providers and arrange and manage their service bookings.

Service Providers
Service providers are electricians, plumbers, cleaners, and other home-service professionals who offer their services through HomeLink. They need a way to make their services visible to potential customers and manage customer service requests.

3.2 Secondary Stakeholders

System Administrator
The system administrator is responsible for the administrative activities required for the operation and management of the HomeLink platform. The exact administrative functions will be refined during the requirements engineering stage.

HomeLink Project/Management Team
The HomeLink project team consists of the students responsible for designing, developing, testing, deploying, maintaining, and improving the system throughout the Software Development Life Cycle.

3.3 Initial Stakeholder Needs

Customer Needs

- Find available home-service providers.
- View relevant information about services and service providers.
- Arrange home-service bookings.
- View and manage their bookings.
- Receive clear information about their service requests.

Service Provider Needs

- Make their services visible to potential customers.
- Provide relevant information about the services they offer.
- Receive customer booking or service requests.
- View and manage received bookings and service requests.

The detailed stakeholder needs will be further analysed and refined during the requirements engineering stage.

4. Project Objectives

4.1 Main Objective

To design and develop HomeLink, a centralized Home Services Booking Platform that conveniently connects customers who require home services with suitable service providers and supports the efficient arrangement and management of service bookings.

4.2 Specific Objectives

- To provide a centralized platform where customers can find suitable home-service providers.
- To enable service providers to present their services and make them accessible to potential customers.
- To facilitate convenient service booking between customers and service providers.
- To enable customers and service providers to manage their service bookings and requests in an organized manner.
- To improve the overall process of connecting customers with home-service providers compared with relying only on personal contacts, recommendations, and separate communication methods.
- To develop a usable, reliable, and secure software solution that supports the core activities of the HomeLink platform.

5. Project Scope

5.1 Scope Statement

HomeLink will be a Home Services Booking Platform designed to connect customers who require household services with service providers such as electricians, plumbers, and cleaners. The system will provide the essential functionality required for customers to discover services and arrange and manage bookings, while enabling service providers to present their services and manage customer service requests. The platform will also provide appropriate administrative functionality for managing the system.

5.2 In Scope

The initial scope of HomeLink includes:

- Customer and service-provider account management.
- Service-provider profiles and service information.
- Different home-service categories, such as electrical, plumbing, and cleaning services.
- Searching or browsing available home services and service providers.
- Viewing relevant service-provider and service details.
- Creating service bookings or service requests.

- Customers viewing and managing their bookings.
- Service providers receiving and managing service requests/bookings.
- Basic administrative functionality required to operate the platform.
- Appropriate input validation.
- User authentication.
- Appropriate authorization according to user roles.
- Error handling.

5.3 Out of Scope

The following features are initially considered outside the scope of the HomeLink mini project:

- Real-time GPS tracking of service providers.
- A dedicated native Android or iOS mobile application.
- Advanced AI-based service-provider recommendations.
- Integration with external smart-home or Internet of Things (IoT) devices.
- Complex business or accounting systems for service providers.
- Nationwide enterprise-level operations and infrastructure.

Features such as online payments, ratings and reviews, notifications, and chat functionality are not finalized at Stage 1. Their inclusion will be considered later during requirements engineering based on stakeholder needs, project priorities, available time, and technical feasibility.

6. Project Assumptions

The development of HomeLink is based on the following initial assumptions.

1. **Internet and Device Access:** Users are assumed to have access to an internet-connected device capable of accessing and using the HomeLink system.
2. **Accurate User Information:** Customers and service providers are expected to provide valid and reasonably accurate information when creating accounts, maintaining profiles, and making service bookings.
3. **Up-to-Date Service Information:** Service providers are expected to maintain and update relevant information about the services they offer.
4. **Sufficient Booking Information:** Customers are expected to provide the necessary information when requesting a home service so that the service provider can understand the service request.
5. **Management of Received Requests:** Registered service providers are expected to review and appropriately respond to service requests or bookings received through HomeLink.
6. **Basic Digital Literacy:** Target users are assumed to have sufficient basic digital knowledge to perform common web-based activities such as registration, login, searching, viewing information, and submitting forms.
7. **Operation Within the Defined Project Scope:** HomeLink is initially being developed as a university Software Engineering mini project rather than as a full-scale commercial home-services platform.

7. Project Constraints

The HomeLink project is subject to several constraints that may influence its development.

- **1. Limited Development Time:** HomeLink must be planned, designed, developed, tested, and deployed within the academic timeframe provided for the Software Engineering mini project.
- **2. Limited Development Resources:** Development will be carried out by the student group using the hardware, software, development tools, and other resources reasonably available to the team.
- **3. Team Knowledge and Experience:** The complexity of the system may be influenced by the team's current knowledge and experience with software development technologies and software engineering practices.
- **4. Project Scope Limitation:** As HomeLink is a university mini project, the team must primarily focus on implementing the prioritized core requirements rather than attempting to reproduce every feature available in large commercial home-service platforms.
- **5. Technology Constraints:** The selected technologies should be suitable for the team's skills, system requirements, available development time, and deployment environment. The technology choices must also be justified in the project documentation.
- **6. Deployment Constraints:** HomeLink must ultimately be deployed in an environment that allows the developed application to be demonstrated to the lecturer. Appropriate configuration and deployment documentation must also be maintained.
- **7. Security and Privacy Considerations:** Since HomeLink will handle user accounts and service-related information, the development team must consider appropriate security and privacy measures when designing and implementing the system.

Process Planning

1. Selected Software Process Model

For the development of **HomeLink – A Home Services Booking Platform**, the team has selected the **Agile Software Development Model**.

Agile is an approach that supports developing a software system through smaller, manageable parts while allowing requirements and priorities to be refined during development. This approach is suitable for HomeLink because the system involves different types of users, mainly customers, service providers, and the system administrator, with different needs and responsibilities.

HomeLink also contains several related features, including user management, service management, service searching, and booking management. These features can be developed and tested gradually instead of waiting until the entire system is completed.

In Stage 1, some possible features such as online payments, ratings and reviews, notifications, and chat functionality were identified but were not finalized. Therefore, the requirements may be further analysed and prioritised during the later stages of the project. Agile provides the flexibility needed to handle such changes without significantly affecting the overall development process.

The main technologies selected for developing HomeLink are:

- **Laravel** – used as the main framework for developing the web application.
- **Docker** – used to provide a consistent development environment for the team.
- **GitHub** – used for source-code version control and team collaboration.

The selected technologies support the team's development process and help maintain consistency and collaboration throughout the project.

2. Justification for the Selected Software Process Model

The Agile model was selected based on the characteristics, requirements, and constraints of the HomeLink project identified in Stage 1.

1. Flexibility in Requirements

Some HomeLink features have not been finalized at the initial project definition stage. For example, the inclusion of online payments, ratings and reviews, notifications, and chat functionality will be considered later based on stakeholder needs and project feasibility.

Agile allows the team to review, refine, and prioritise requirements as the project progresses. Therefore, changes to requirements can be handled more easily.

2. Suitable for Incremental Development

HomeLink contains several related modules, such as user management, service management, service searching, and booking management. These modules do not need to be developed at the same time.

The team can develop the core modules in smaller increments and gradually integrate them into the complete system. This makes the development work more manageable for the team.

3. Early Testing and Improvement

Agile supports testing throughout development rather than leaving all testing until the end. After completing an increment, the team can test the implemented features and identify errors or missing requirements.

This allows problems to be corrected earlier and helps improve the quality of the HomeLink system before the final deployment.

4. Suitable for the Five-Member Team

HomeLink is being developed by a five-member student team. The project contains several modules and activities that can be divided into manageable tasks among team members.

Agile supports regular communication and collaboration, allowing team members to coordinate their work and contribute to different stages of development. This also supports the project guideline requiring each member to make a meaningful contribution to the project.

5. Suitable for Limited Time and Resources

As HomeLink is a university Software Engineering mini project, the team has limited development time and resources. Therefore, it is important to focus on the most important features first.

Using Agile, the team can prioritise the core functionality, such as user management, service management, service searching, and booking management, before considering additional features.

6. Supports Continuous Improvement

The team can review the work completed during each development cycle and identify areas that need improvement. Feedback from testing and reviews can then be used to improve the next increment.

Therefore, Agile is considered appropriate for HomeLink because it provides flexibility, incremental development, early testing, teamwork, and continuous improvement while supporting the project's limited time and resources.

3. Project Plan

The HomeLink project will follow the SDLC stages specified in the ICT2232 mini project guidelines. Each stage will produce outputs that provide a foundation for the following stage. This helps the team maintain a clear connection between the problem identified in Stage 1 and the final software solution.

SDLC Stage	Main Activities	Main Output
Stage 1 – Problem & Project Definition	Identify the problem, stakeholders, objectives, scope, assumptions and constraints	Project Proposal
Stage 2 – Process Planning	Select and justify the software process model and define the development approach	Process Plan
Stage 3 – Agile Planning	Prepare product backlog, user stories, priorities, acceptance criteria and sprint plan	Agile Plan
Stage 4 – Requirements Engineering	Identify, analyse and prioritise functional and non-functional requirements	Requirements Specification
Stage 5 – System Design	Design system architecture, database, UML models and user interfaces	Design Document
Stage 6 – Implementation	Develop and integrate the prioritised HomeLink features	Working System
Stage 7 – Testing	Prepare test cases, perform testing and correct identified defects	Test Results
Stage 8 – Deployment	Configure and deploy the application	Deployed System
Stage 9 – Maintenance	Analyse and implement the approved change request	Updated System
Stage 10 – Security	Identify security threats and implement appropriate security controls	Security Assessment

3.1 Planned Development Priority

Based on the scope identified in Stage 1, the initial development priority for HomeLink will be as follows.

1. User Management

- Customer and service-provider registration and login
- User profiles
- User roles

This will provide the basic user accounts and access control required for the other system functions.

2. Service Management

- Service categories
- Service-provider profiles
- Service information
- Managing offered services

This will allow service providers to present their available services through the platform.

3. Service Searching

- Browse service categories
- Search for services
- View service-provider details

This will allow customers to find relevant services and service providers.

4. Booking Management

- Create service bookings
- View bookings
- Receive booking requests
- Manage booking requests

This will support the main purpose of HomeLink, which is to connect customers with service providers and manage service bookings.

5. Integration and Final Improvements

- Administrative functions
- Input validation
- Error handling
- Security
- Integration of major modules

The priorities may be adjusted during Agile Planning and Requirements Engineering based on the finalised and prioritised requirements.

4. Development Approach

The team will use an iterative and incremental development approach within the selected Agile model.

In an incremental approach, the system is developed and delivered through smaller functional parts. In an iterative approach, the developed parts can be reviewed, tested, and improved based on the findings from each development cycle.

For HomeLink, these approaches will be combined. Instead of attempting to develop the complete platform at once, the team will first focus on the core functionality and gradually add and improve other features.

Each development cycle will generally follow:

Planning → Design → Implementation → Testing → Review and Improvement

The result of one cycle will be used to guide the work of the next cycle.

4.1 Development Increments

Increment	Main Focus
Increment 1	User registration, login, profiles and roles
Increment 2	Service-provider and service management
Increment 3	Service searching and viewing service details
Increment 4	Service booking and booking management
Increment 5	Administration, validation, security, integration and final improvements

This order gives priority to the core functions required for the HomeLink platform. For example, users and service providers must be able to register before services can be managed, and services need to be available before customers can search for and book them.

4.2 Development Technologies

Laravel

Laravel will be used as the main framework for developing the HomeLink web application. It will provide the main structure required to implement the application's features and manage communication between the user interface, application logic, and database.

Docker

Docker will be used to maintain a consistent development environment for all five team members. This can reduce differences between individual environments and make it easier for team members to run the project using similar configurations.

GitHub

GitHub will be used for source-code version control and collaboration. Team members can maintain their work in a shared repository, track changes, and integrate their contributions. The repository history will also provide evidence of individual contributions as required by the project guidelines.

4.3 Overall Development Flow

The overall development approach can be summarised as:

Identify and Prioritise Requirements
↓
Plan an Increment
↓
Design
↓
Develop
↓
Test
↓
Review and Improve
↓
Continue with the Next Increment

This approach allows the team to gradually build HomeLink, identify problems early, and make improvements while keeping the development work manageable within the available academic timeframe.

Agile Planning

1. Introduction to Agile Approach

For the HomeLink platform, the team has selected the Agile Software Development Model, continuing from the process planning stage. Agile allows the team to develop the system through manageable increments and refine user requirements as the project progresses. This approach is ideal for HomeLink because it accommodates different user roles such as customers, service providers, and administrators — and supports iterative adjustments to features like booking management and service searching. Rather than attempting to design and build the entire platform at once, the team will work through a series of short, focused development cycles, each delivering a usable set of features that can be reviewed and refined before moving on to the next. This reduces the risk of major rework later in the project and allows early feedback to be incorporated while there is still time to act on it. Agile also supports better coordination within the five-member team, since work can be broken down into smaller, clearly defined tasks that are easier to plan, assign, and track through regular check-ins. Given the limited timeframe of the mini project, this incremental and iterative approach helps the team prioritise the most important functionality first while still leaving room to accommodate features that were not fully finalised during Stage 1, such as payments, ratings, notifications, and chat.

2. Product Backlog

Backlog ID	User Story Summary	Priority	Estimate (Story Points)
PB-01	As a user, I want to register and log into my account with appropriate roles (customer, service provider, administrator) so that I can securely access the platform.	Must Have	5
PB-02	As a service provider, I want to create and manage my profiles, service categories, and offerings so that customers can discover them.	Must Have	5
PB-03	As a customer, I want to browse service categories and search for available home-service providers so that I can find necessary professionals.	Must Have	5
PB-04	As a customer, I want to create and track service bookings, and as a provider, I want to receive and manage incoming booking requests.	Must Have	8
PB-05	As an administrator, I want basic management tools to oversee system operations and user accounts.	Should Have	3
PB-06	As a user, I want proper input validation and error handling across forms to ensure system reliability.	Must Have	3

3. Detailed User Stories and Acceptance Criteria

User Story 1: User Authentication & Role Assignment

User Story ID: US-01

As a customer or service provider,
I want to register and log into the system securely,
So that I can access my designated role-specific dashboard**.**

Acceptance Criteria:

- Users can fill out and submit registration and login forms with valid details.
- The system authenticates credentials and routes users according to their assigned role (customer or service provider).
- Proper error messages appear if invalid data or incorrect credentials are entered.

User Story 2: Service Provider Profile Management

User Story ID: US-02

As a service provider,
I want to update my profile and manage offered services,
So that potential customers can view accurate business information.

Acceptance Criteria:

- Providers can select service categories (such as electrical, plumbing, or cleaning).
- Providers can edit their service descriptions, pricing notes, and availability status.

User Story 3: Searching for Home Services

User Story ID : US-03

As a customer,
I want to browse service categories and search for providers,
So that I can choose a suitable professional.

Acceptance Criteria:

- Customers can filter providers based on specific home service categories.
- Selecting a listing displays full provider details and offered services.

User Story 4: Booking Management

User Story ID: US-04

As a customer,
I want to submit a service booking request and track its status,
So that I can arrange household maintenance conveniently.

Acceptance Criteria:

- Customers can input booking details (preferred date, time, and service type) and submit requests.
- Service providers can view incoming requests on their dashboard and accept or reject them.

4. Sprint Planning & Release Increments

Development aligns with the five incremental phases established during process planning:

- **Increment 1:** User registration, login, profiles, and role management.
- **Increment 2:** Service-provider profile setup and service catalog management**.**
- **Increment 3:** Service searching, filtering, and viewing provider details.
- **Increment 4:** Service booking creation and booking request management.
- **Increment 5:** Administration, input validation, security controls, module integration, and final Testing**.**

5. Task Allocation and Team Coordination

- **Team Structure:** Developed collaboratively by the five-member student group.
- **Tooling:** Built using Laravel for backend logic, Docker for containerized environment consistency, and GitHub for version control and task tracking.
- **Workflow:** Team tasks are managed through iterative sprints, with code changes integrated continuously into GitHub repositories to maintain a clear individual contribution history.

Requirements Engineering

Requirements Engineering

1. Stakeholder Needs

Building on the stakeholders identified in Stage 1, their specific needs are analysed below to form the basis for the requirements.

Customer Needs

- Register and securely access an account.
- Browse and search for home-service providers by category.
- View detailed information about a provider and their services before booking.
- Create a service booking with relevant details (date, time, service type).
- Track the status of submitted bookings.
- Receive clear feedback when input is invalid or an action fails.

Service Provider Needs

- Register and securely access an account.
- Create and maintain a profile with service categories, descriptions, pricing notes, and availability.
- Receive booking requests from customers.
- Accept or reject incoming booking requests.
- Manage which services are currently visible to customers.

System Administrator Needs

- View and manage registered user accounts (customers and providers).
- Disable or remove accounts that violate platform rules.
- Have basic oversight of platform activity.

HomeLink Project Team Needs

- A requirements set that is clear, prioritised, and traceable through design, implementation, and testing.
- Requirements scoped appropriately for the team's time and resources.

2. Functional Requirements

ID	Functional Requirement	Related Stakeholder(s)
FR-01	The system shall allow customers and service providers to register an account with role selection (customer or service provider).	Customer, Provider
FR-02	The system shall allow registered users to log in using valid credentials.	Customer, Provider
FR-03	The system shall authenticate users and redirect them to a role-specific dashboard after login.	Customer, Provider

FR-04	The system shall allow users to view and update their own profile information.	Customer, Provider
FR-05	The system shall allow service providers to create service listings under predefined categories (e.g., electrical, plumbing, cleaning).	Provider
FR-06	The system shall allow service providers to edit or remove their service listings, including descriptions, pricing notes, and availability status.	Provider
FR-07	The system shall allow customers to browse available service categories.	Customer
FR-08	The system shall allow customers to search for service providers using filters such as category.	Customer
FR-09	The system shall display detailed service-provider and service information when a listing is selected.	Customer
FR-10	The system shall allow customers to create a service booking request, including preferred date, time, and service type.	Customer
FR-11	The system shall allow customers to view the status of their submitted bookings.	Customer
FR-12	The system shall allow service providers to view incoming booking requests on their dashboard.	Provider
FR-13	The system shall allow service providers to accept or reject booking requests.	Provider
FR-14	The system shall allow administrators to view all registered user accounts.	Administrator
FR-15	The system shall allow administrators to disable or remove a user account.	Administrator
FR-16	The system shall validate user input on all forms (registration, profile, service listing, booking) and display appropriate error messages for invalid input.	All
FR-17	The system shall restrict access to functionality based on the user's assigned role (customer, provider, administrator).	All

3. Non-Functional Requirements

ID	Category	Non-Functional Requirement
NFR-01	Performance	The system shall load core pages (search results, dashboards) within 3 seconds under normal load.
NFR-02	Usability	The system shall provide a simple, consistent interface usable by users with basic digital literacy, requiring no special training.
NFR-03	Reliability	The system shall handle invalid operations (e.g., failed submissions) gracefully without crashing, returning clear error messages.
NFR-04	Security	The system shall store user passwords using secure hashing and restrict data access based on authenticated user roles.
NFR-05	Maintainability	The system shall be developed using a modular Laravel architecture so that individual modules (user, service, booking) can be updated independently.
NFR-06	Availability	The deployed system shall be accessible to users whenever the hosting environment is running, with minimal planned downtime.
NFR-07	Portability	The system shall run consistently across development and deployment environments through the use of Docker containerisation.

4. Use Cases / Use Case Descriptions

Actors: Customer, Service Provider, Administrator

Identified Use Cases:

- Register Account
- Login
- Manage Profile
- Manage Services (Provider)
- Browse/Search Services (Customer)
- View Provider Details (Customer)
- Create Booking (Customer)
- Manage Booking Requests (Provider)
- View Booking History (Customer)
- Manage User Accounts (Administrator)

Use Case Descriptions

UC-01: Register Account

- **Actor:** Customer, Service Provider
- **Precondition:** User does not already hold an account.
- **Main Flow:** User selects "Register," enters details and selects a role, and submits the form; system validates the input and creates the account.
- **Postcondition:** A new user account is created and the user can log in.

UC-02: Login

- **Actor:** Customer, Service Provider, Administrator
- **Precondition:** User holds a registered account.
- **Main Flow:** User enters credentials; system authenticates and redirects to the role-specific dashboard.
- **Postcondition:** User is logged in and has access to role-appropriate functions.

UC-03: Manage Services

- **Actor:** Service Provider
- **Precondition:** Provider is logged in.
- **Main Flow:** Provider creates, edits, or removes a service listing, including category, description, pricing notes, and availability.
- **Postcondition:** Service listing is created/updated and visible to customers (if active).

UC-04: Browse/Search Services

- **Actor:** Customer
- **Precondition:** None (browsing does not require login).
- **Main Flow:** Customer selects a category or enters search criteria; system returns matching providers.
- **Postcondition:** Customer views a list of relevant providers.

UC-05: View Provider Details

- **Actor:** Customer
- **Precondition:** Customer has performed a search or browse.
- **Main Flow:** Customer selects a listing; system displays full provider and service details.
- **Postcondition:** Customer has the information needed to decide whether to book.

UC-06: Create Booking

- **Actor:** Customer
- **Precondition:** Customer is logged in and viewing a provider's details.
- **Main Flow:** Customer enters preferred date, time, and service type and submits the request; system validates input and creates the booking with "Pending" status.
- **Postcondition:** Booking is created and appears on the provider's dashboard.

UC-07: Manage Booking Requests

- **Actor:** Service Provider
- **Precondition:** Provider is logged in and has received at least one booking request.
- **Main Flow:** Provider reviews a request and accepts or rejects it; system updates the booking status.

- **Postcondition:** Booking status is updated and visible to the customer.

UC-08: View Booking History

- **Actor:** Customer
- **Precondition:** Customer is logged in and has made at least one booking.
- **Main Flow:** Customer opens "My Bookings"; system displays past and current bookings with status.
- **Postcondition:** Customers can review their booking history.

UC-09: Manage User Accounts

- **Actor:** Administrator
- **Precondition:** Administrator is logged in.
- **Main Flow:** Administrator views the list of registered users and disables/removes a selected account if necessary.
- **Postcondition:** User account list is updated accordingly.

5. Requirements Prioritisation (MoSCoW)

ID	Requirement Summary	Priority
FR-01	Registration with role selection	Must
FR-02	Login	Must
FR-03	Role-based redirect after login	Must
FR-04	Profile view/update	Should
FR-05	Create/manage service listings	Must
FR-06	Edit/remove service listings	Should
FR-07	Browse service categories	Must
FR-08	Search with filters	Must
FR-09	View provider/service details	Must
FR-10	Create booking request	Must
FR-11	View booking status	Must
FR-12	Provider views incoming requests	Must
FR-13	Accept/reject booking requests	Must
FR-14	Admin views user accounts	Should
FR-15	Admin disables/removes accounts	Could
FR-16	Input validation & error handling	Must

FR-17	Role-based access control	Must

This is consistent with the Stage 3 product backlog priorities (PB-01 to PB-06), where core account, service, search, and booking functions were rated "Must Have," and administrative tools were rated "Should Have"/"Could Have" given their lower urgency relative to core booking functionality.

6. Requirements Traceability

The table below links each functional requirement to the user story it originates from, the use case that describes its behaviour, and the development increment (from Stage 2/3) in which it will be implemented. Design artefact references (e.g., diagram/component names) and test case IDs will be added once Stage 5 (Design) and Stage 7 (Testing) are completed, extending this same matrix rather than creating a new one — this preserves a single, continuous traceability record across the SDLC.

Req. ID	Source User Story	Related Use Case	Planned Increment	Design Artefact (Stage 5)	Test Case ID (Stage 7)
FR-01	US-01	UC-01	Increment 1	*To be added*	*To be added*
FR-02	US-01	UC-02	Increment 1	*To be added*	*To be added*
FR-03	US-01	UC-02	Increment 1	*To be added*	*To be added*
FR-04	US-02, US-09	UC-02	Increment 1	*To be added*	*To be added*
FR-05	US-02	UC-03	Increment 2	*To be added*	*To be added*
FR-06	US-02, US-09	UC-03	Increment 2	*To be added*	*To be added*
FR-07	US-03	UC-04	Increment 3	*To be added*	*To be added*
FR-08	US-03	UC-04	Increment 3	*To be added*	*To be added*
FR-09	US-03, US-08	UC-05	Increment 3	*To be added*	*To be added*
FR-10	US-04	UC-06	Increment 4	*To be added*	*To be added*
FR-11	US-04, US-11	UC-08	Increment 4	*To be added*	*To be added*
FR-12	US-05	UC-07	Increment 4	*To be added*	*To be added*
FR-13	US-05	UC-07	Increment 4	*To be added*	*To be added*
FR-14	US-06	UC-09	Increment 5	*To be added*	*To be added*
FR-15	US-12	UC-09	Increment 5	*To be added*	*To be added*
FR-16	US-07	All	Increment 5	*To be added*	*To be added*
FR-17	US-10	All	Increment 5	*To be added*	*To be added*